

RESEARCH PLAN



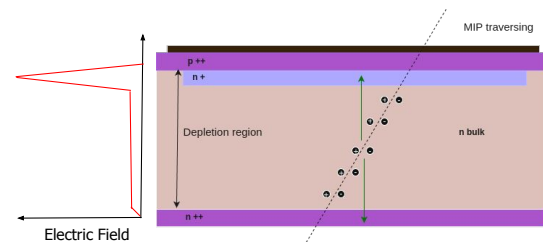
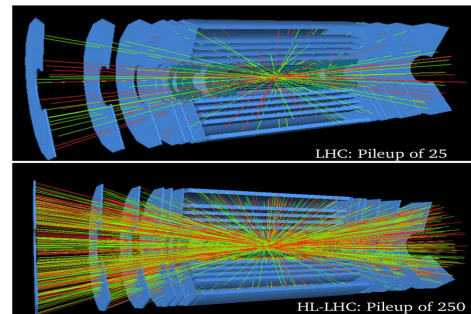
Radiation Damage Modelling of 4H-SiC Low Gain Avalanche Detectors (LGADs) Using Silvaco TCAD: Verification Against Multi-Foundry Experimental Data and Further Studies

Jaideep Kalani

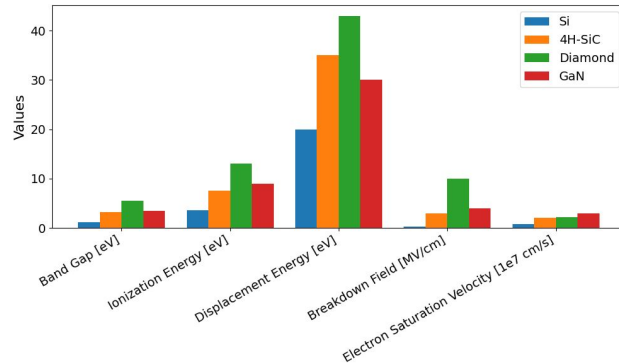
School of Physical Sciences
(IIT Mandi)
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Motivation

- HL-LHC needs ultra-fast timing detectors with picosecond precision
- Detectors must survive very high radiation levels
- Si-LGADs are current standard for 4D tracking in CMS and ATLAS
- Performance degrades due to Acceptor Removal Mechanism and Defects
- Si-Boron gain layer weakens with irradiation
- Leads to higher voltage needs and cooling challenges
- 4H-SiC offers strong radiation hardness
- Wide bandgap reduces leakage current even after damage
- Can potentially operate at room temperature
- Early 4H-SiC LGAD prototypes already demonstrated
- No complete TCAD radiation damage model available yet
- This work builds and validates a physics-based model in Silvaco TCAD
- Then uses it to predict and optimise new detector designs



Schematic structure of a typical p-in-n LGAD



Gap Analysis

Current researchs	What is missing
Baseline 4H-SiC LGAD IV/CV and charge collection with RASER simulation	Any simulation in Silvaco TCAD
Proton damage model in Sentaurus (SICAR [2])	Full IV , CV characteristics with proton and neutron irradiation defect model in Silvaco with doping removal
Neutron damage model for 4H-SiC PIN diodes in GTS-TCAD (Gaggl et al. [6,7])	Neutron damage model for 4H-SiC LGAD in Silvaco
First-order TCAD checks by NCSU/LBNL/BNL [3]	Proper calibrated proton and neutron damage model for NCSU/LBNL/BNL structure verification in TCAD (for LGAD)
Temperature-dependent IV in Onsemi 4H-SiC LGAD (Novotný et al. [4], [5])	Verification with TCAD, Radiation damage and fluence study on the structure
Proton vs neutron irr. comparison for Si LGAD well established	Proton vs neutron irr. comparison for 4H-SiC LGAD

Production Facilities

Source of Production	Device Type	References
NCSU (USA, in-house)	LGAD + PiN	<i>Impact of Proton Irradiation on 4H-SiC LGADs</i> , arXiv:2507.23062
IMB-CNM (Spain, foundry)	PiN	<i>TCAD modeling of radiation-induced defects in 4H-SiC diodes</i> , NIMA
IMB-CNM (Spain, foundry)	LGAD (trench design)	<i>Design and Simulation of a 4H-SiC LGAD with Trench-Isolation</i> , arXiv:2510.14531
IHEP China (in-house, SICAR)	LGAD	<i>The study of 4H-SiC LGAD after proton radiation</i> , arXiv:2507.12238
ON Semiconductor (ONSEMI) (with CTU/FZU)	LGAD	<i>First generation 4H-SiC LGAD production and its performance evaluation</i> , arXiv:2503.07490

Research strategy

- Learn using Silvaco TCAD, code structure, deckbuild, plotting, etc.
- Try to make baseline PiN and LGAD structures, IV characteristics, mesh defining, convergence, 2D/3D geometry, doping, dimensions, etc.
- Temperature dependence study of LGAD gain, comparison with experimental data
- Time resolution / Gain comparison of PiN and LGAD and compare with exp. data
- Making and incorporating damage models for neutron and proton irradiation
- Accpetor removal models
- Proton vs neutron damage model comparison
- Radiation hardness studies
- Novel work like ultra thin LGAD, JTE for delayed breakdown, 2D map of (bias, temperature) that maximises gain post-irradiation, Gain layer thickness trade-off, Compensated gain layer concept for SiC

Experimental Reference Datasets and Device Structures

References

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